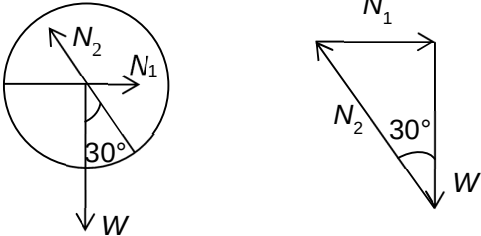


		$R_x = 107 \text{ N (to the right)}$
6	C	

Qn	Ans	Suggested Solution
		Resolving the forces, $N_2 \sin 30 = N_1 \quad (1)$ $N_2 \cos 30 = W \quad (2)$ $\frac{(1)}{(2)}: \quad \tan 30 = \frac{N_1}{W}$ $N_1 = W \tan 30 = 15 \tan 30 = 8.66 = 8.7 \text{ N (2s.f.)}$ (Alternative: Using vector triangle) $N_1 = W \tan 30 = 15 \tan 30 = 8.66 = 8.7 \text{ N (2s.f.)}$
7	A	By conservation of energy, elastic potential energy is converted to kinetic energy and work